# Radu Briciu

BSc, PGDip

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## Education

**Postgraduate Diploma in Quantitative Finance** — City of London, United Kingdom
*Bayes Business School (formerly Cass) – City, University of London* — *Sept. 2023 — Jun. 2025*

**Bachelor of Science in Finance (with honours)** — Westminster, United Kingdom
*Westminster Business School – University of Westminster* — *Aug. 2019 — Jun. 2023*

## Experience

**Finance & Operations Intern** — London, United Kingdom
*Novo Nordisk UK Ltd* — *Aug. 2021 — Aug. 2022*

a. Completed a 12-month placement split between the 'Commercial Excellence' and 'Business Finance Analytics and Pricing' teams
b. Built 'Sales vs Target' and cost variance models and automated periodic reporting tasks
c. Collected, synthesized, and presented analytics on market dynamics, brand performance, and customer identification
d. Verified and consolidated financial accounts and supported analysis of financial performance
e. Cleaned CRM data and collected competitor price data

**Intern** — Bucharest, Romania
*Bristol-Myers Squibb* — *Jun. — Jul. 2018*

a. Managed databases, documented archives, and performed basic data analysis
b. Consolidated notes and planning documentation

## Projects

**Machine Learning Option Pricing** — `github.com/boomelage/machine-learning-option-pricing`
Distillation of option pricing models using feedforward neural networks pricing barrier and average-price options on Heston-calibrated SPX surfaces, trained on large market calibrated parameters and priced with QuantLib; basis of the publication below.

**Heston Simulation** — `github.com/boomelage/heston-sim`
Semi-analytical simulation scheme for the Heston stochastic volatility model, using exact non-central chi-squared variance sampling and a trapezoidal integrated-variance update, with a plain NumPy implementation.

**Binomial Option Pricing** — `github.com/boomelage/bopm`
Derivation of a closed-form expression for the Cox–Ross–Rubinstein binomial model, collapsing the backward lattice recursion into a single discounted binomial sum over terminal payoffs.

**Realised Volatility** — `github.com/boomelage/realised-volatility`
A statistical enquiry into the measurement of realised volatility, fitting ARMA–GARCH models to high-frequency returns and comparing forecast accuracy across specifications.

**Efficient Frontier Optimization** — `github.com/boomelage/ef-optimization`
Markowitz mean–variance portfolio allocation in Python with constrained quadratic optimization and efficient-frontier visualization across the risk–return trade-off.

**prettymath** — `github.com/boomelage/prettymath`
LaTeX package wrapping theorem-like statements in colour-coded, cross-referenced boxes, with configurable environments for definitions, theorems, and proofs in mathematical notes.

**Web Tetris** — `free-tetris.online`
Hobby horse browser Tetris with levels, line-clear scoring, piece hold, and ghost preview, written in dependency-free vanilla HTML/CSS/JavaScript.

## Publications

Briciu, Radu (Sept. 2024). "Neural Networks for Exotic Option Pricing". DOI: 10.2139/ssrn.5104328.

## Skills

Machine Learning – `Python` (QuantLib, NumPy, Pandas, sklearn, keras) – MATLAB – LaTeX
CLI (PowerShell, Bash, WSL, Git, Github) – Editors (Sublime Text, Vim, VSCode, TeXstudio, JupyterLab)
CLI Agents (Claude Code, Codex, Gemini) – Bloomberg Terminal (VCUBE, OMON, Historical Data) – Ollama
AlphaVantage API – BIOS/OS Configuration – MS Office & Power Query

## Courses

Advanced Corporate Finance – Applied Machine Learning – Asset Pricing – Blockchain, Cryptocurrencies and Decentralized Finance – Corporate Finance – Corporate Financial Accounting – Derivatives – Development Economics – Econometrics of Financial Markets – Financial Modelling and Statistics – Fixed Income – Foundations of Econometrics – Global Business Environment – Introduction to Finance – Introduction to Financial Markets – Investments and Portfolio Management – Law and the Business – Mathematics for Finance – Money Banking and Financial Markets Numerical Methods: Applications – Polylang German 1 – Professional and Research Skills for Finance – Programming for Quants – Risk Analysis Risk Management – Securities Analysis – Stochastic Modelling Methods in Finance – Strategic Perspectives for Finance and Accounting